

Ragib Sakib

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EDUCATION

University of Southern California | Los Angeles, CA | *August 2026 - June 2028 (Expected)*

Master of Science (MS) in **Computer Science (Artificial Intelligence)**

University of California, Irvine | Irvine, CA | *September 2020 - June 2024*

Bachelor of Science (BS) in **Software Engineering**, GPA: **3.6**

TECHNICAL SKILLS

Programming Languages: Python, C++, C#, Rust, Java

Data & AI: PyTorch, TensorFlow, OpenCV, FAISS, sentence-transformers, ChromaDB, SQL, SQLite, Matplotlib, Scipy

Web Development: React, TypeScript, Angular, HTML, CSS, JavaScript, Node.js, Ionic, React Native, Gradio

Tools & Platforms: Docker, Jira, Linux, MacOS, Windows

WORK EXPERIENCE

Canon USA

Software and Machine Learning Engineer (Trainee) | Irvine, CA | *September 2023 - February 2025*

- Prototyped a Python pipeline using SAM 2 segmentation, Hough transforms, and clustering for object corner localization
- Ported the spatial clustering and geometric pipelines to C++, reducing execution latency by 90% via pass-by-reference
- Integrated the finalized model runtimes into production software, enhancing user experience via low-latency interactivity
- Shipped an app built in Ionic React to both IOS and Android platforms using one codebase to expand user reach

Canon USA

Software Intern | Irvine, CA | *June 2023 - August 2023*

- Converted major C# codebase from old UWP standard to current standard to remove technical debt
- Created a multi-platform mobile and web application that expanded upon Project Team's current web-only client
- Operated within an Agile schedule work environment while making full use of Jira

Boeing Satellite Systems

Engineering Intern | El Segundo, CA | *June 2019 - August 2019*

- Led a quality engineering team in an intern competition before 300 engineers, placing top three in every category
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SOFTWARE PROJECTS

LoL Team Recommender | *September 2026*

DEMO/Write-up link: <https://github.com/RSakib/LoL-Team-Optimizer---Two-Tower-RecSys-and-RAG>

- Trained a PyTorch two-tower model on 69,378 player pairs to rank teammates by role, rank, and champion preferences
- Built a joint lineup scorer using player interactions and champion pools to optimize four teammate selections together
- Integrated Sentence Transformers, ChromaDB, and Qwen RAG to explain recommendations using 2,854 real player profiles
- Deployed a Dockerized Gradio/FastAPI app on Google Cloud Run with scale-to-zero settings to minimize idle hosting costs
- Benchmarked role-and-rank retrieval to validate ranking, exceeding popularity NDCG@10 by 29%

MovieLens 1M Two-Tower Recommendation Engine | *June 2026*

DEMO/Write-up link: <https://github.com/RSakib/Pre-2001-Movie-Recs-via-Two-Tower-Neural-Network>

- Built a two-tower neural retrieval model with in-batch negative ranking loss to provide movie recommendations
- Engineered user tower encoding gender, age group, and sequential watch history to capture personalized preference signals
- Enabled fast recommendation retrieval at inference time by indexing movie catalog into FAISS flat inner product index
- Generated t-SNE visualization confirming genre-based clustering in learned embedded space

MUSIC Gait Analysis Software (College Capstone) | *September 2023 - June 2024*

Project Lead/Front-End Engineer and Designer

- Led a team of 3 programmers and 2 designers to create a cross-platform rehabilitation app for an Irvine-based start-up
- Application connected to rehabilitation motion sensor to provide meaningful feedback on user's walk gait
- Standardized code format with python linter, created unit tests for 80% of our code, and fully documented our design